

ADITYA KULKARNI

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EDUCATION

North Carolina State University

M.S. Computer Science

Raleigh, NC

Aug 2024 – May 2026

University of Mumbai

B.Tech. Computer Science, CGPA: 9.32/10

Mumbai, India

Aug 2020 – May 2024

WORK EXPERIENCE

Product Engineer Intern

May 2025 – Present

Isomer AI

Remote

- Shipped 10+ production backend features for an AI-powered P&C claims automation platform – Temporal.io workflow orchestration, RBAC, and real-time event pipelines – integrated with Guidewire via REST API calls and export parsing, across 7 deployments using TypeScript, Node.js, Express, PostgreSQL, and Redis
- Built document processing pipelines (PDF generation, email parsing via Microsoft Graph API, S3 storage) and multi-model LLM tooling across Claude, GPT-4o, and Gemini with MCP; contributed 394 commits and led code review with AI-assisted development (Claude Code, Copilot)

Software Developer Intern

Jun 2024 – Aug 2024

Mphasis (HP Company)

Remote

- Built 5+ AI-driven automation workflows including an OCR and NLP-powered insurance claim processing pipeline, improving accuracy 25% in production; automated CI/CD on AWS and GCP with Grafana monitoring dashboards

Graduate Research Assistant

Apr 2025 – Present

NC State University

Raleigh, NC

- Engineered a cloud-based genomic data platform on Google Cloud Run with JBrowse 2, enabling interactive visualization of 500GB datasets for 100+ researchers; improved ML model accuracy 12% via hyperparameter tuning and time-series analysis

PERSONAL PROJECTS

KubeServe – Kubernetes LLM Inference Platform | Python, FastAPI, Kubernetes, KEDA, Helm, ArgoCD, Otel 2026

- Built a Kubernetes-native LLM inference platform: Python Kopf CRD operator reconciling Deployments and KEDA ScaledObjects; FastAPI router with 3-state circuit breaker and EWMA load balancing across Claude, GPT-4o, and Gemini
- OpenTelemetry + Prometheus + Grafana full observability stack; k6 load-tested to 500 VUs; 88 pytest tests; Helm + ArgoCD GitOps; multi-stage Docker builds with scale-to-zero on idle

PulseRank – Real-Time Leaderboard & Activity Feed | TypeScript, NestJS, Next.js, MySQL, MongoDB, Redis 2026

- Built an event-driven leaderboard and feed platform: MySQL for transactional writes with idempotency-key deduplication and outbox pattern, MongoDB for feed projections with TTL expiry, Redis sorted sets for sub-millisecond reads via Server-Sent Events
- Validated multi-store consistency with Testcontainers integration tests; benchmarked hot paths with k6; OpenTelemetry + Prometheus observability; GitHub Actions CI/CD

Inventory Reservation Service | Java 21, Spring Boot 3.5, PostgreSQL, Testcontainers, Docker 2026

- Built a transactional inventory reservation REST API in Java 21 and Spring Boot preventing overselling via PostgreSQL row-level locking (SELECT FOR UPDATE) with idempotent endpoints and RFC 9457 error responses
- Verified concurrency correctness with Testcontainers – two concurrent requests against quantity 1 produce exactly one success and one 409 Conflict; Flyway migrations, Spring Security RBAC, multi-stage Docker, GitHub Actions CI

TECHNICAL SKILLS

Languages: TypeScript, Java, Python, JavaScript, SQL, R, C

Full Stack: Node.js, Express, NestJS, React, Next.js, Spring Boot, FastAPI, REST APIs, GraphQL

Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch

Infra & Observability: Docker, Kubernetes, AWS, GCP, Terraform, GitHub Actions, OpenTelemetry, Prometheus, Grafana